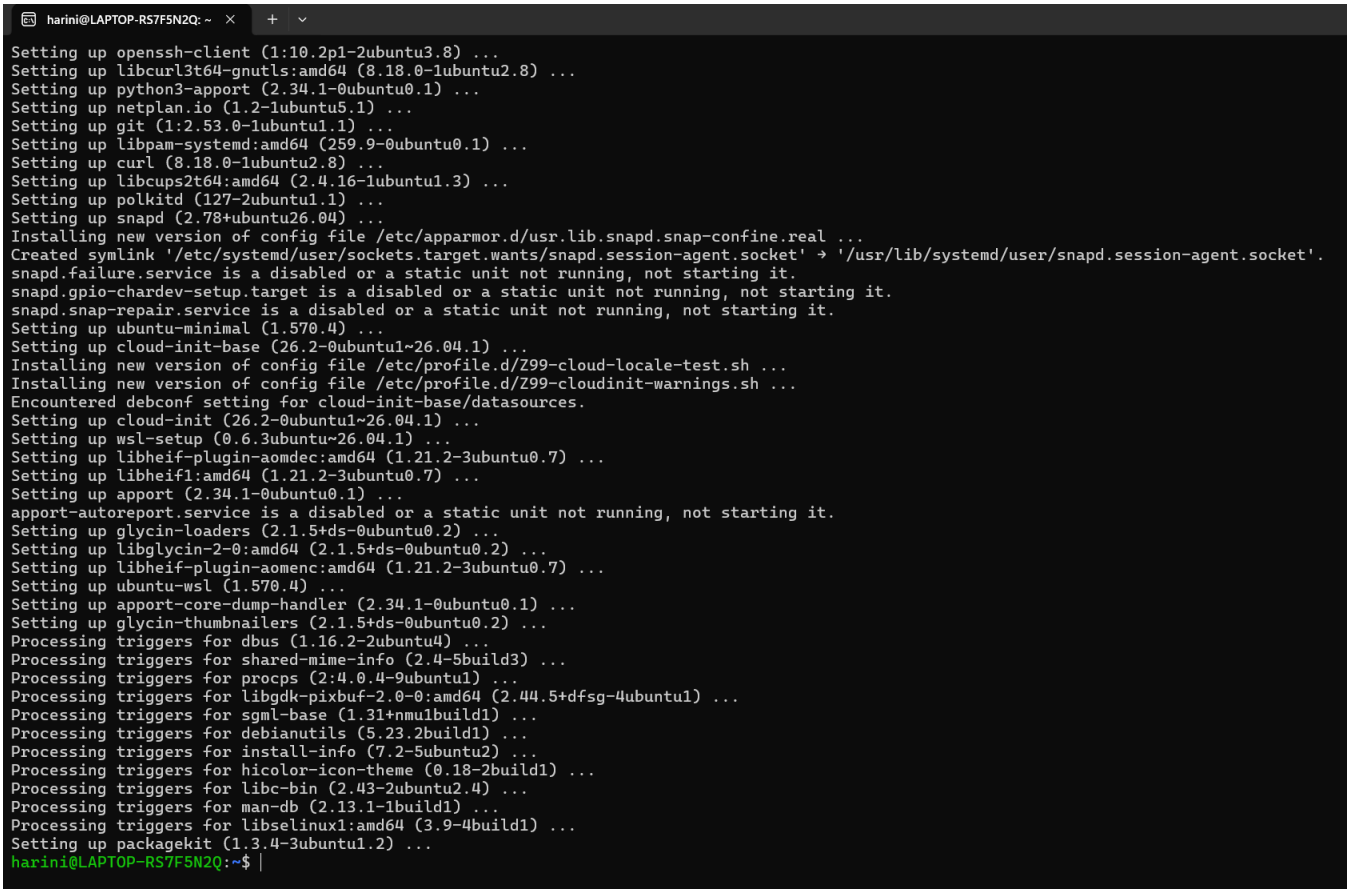
CNNP- Skill 2

THREE-POINT CAPTURE – Analyze packet captures from three points in a network path to understand packet flow and identify performance differences.

STAGE 1 — Install tools

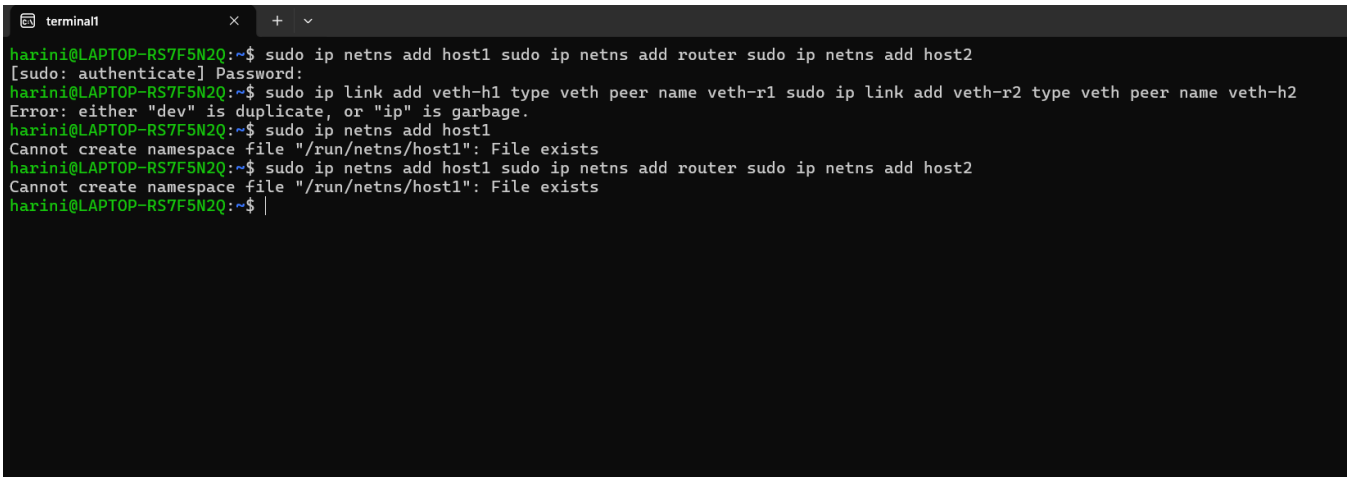
Open Ubuntu and type the following commands in the Ubuntu terminal

sudo apt update
sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd



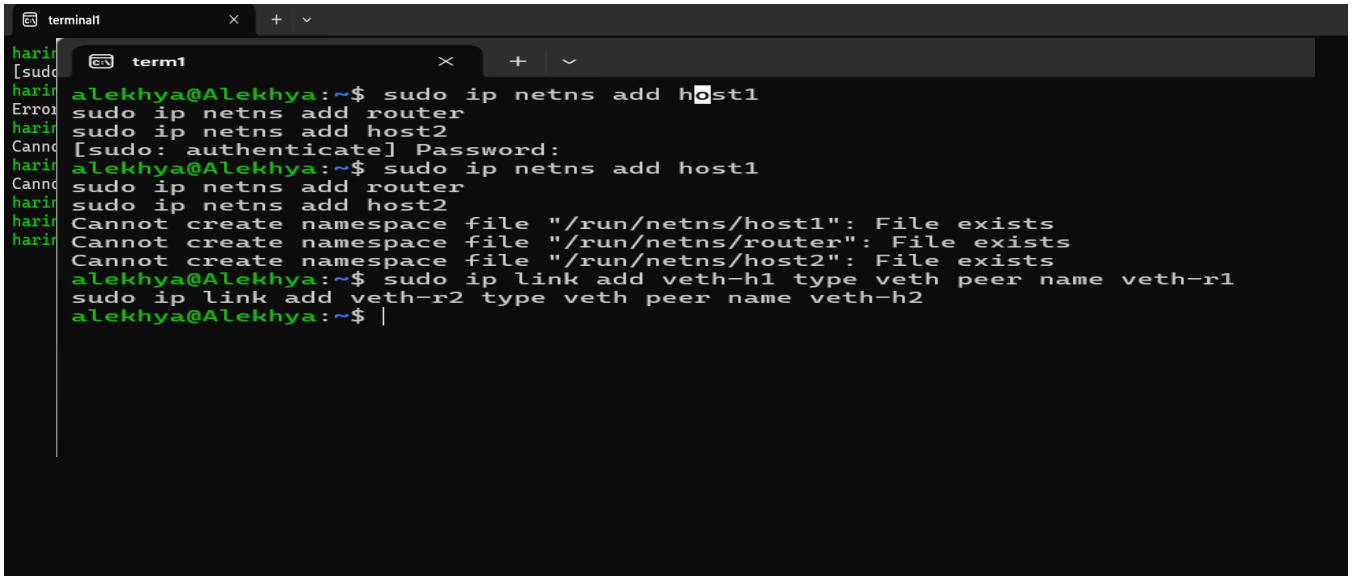
STAGE 2 — Build the network

Open again ubuntu and rename that tab as Terminal 1 and run all of these one by one:



sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2

sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2



sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router

sudo ip link set veth-h2 netns host2

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns del host1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link del veth-h2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns list
host2
router
host1
harini@LAPTOP-RS7F5N2Q:~$
```

STAGE 3 — Configure IP addresses (Note: In terminal 1 Only)

host1:

sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1

sudo ip netns exec host1 ip link set veth-h1 up

sudo ip netns exec host1 ip link set lo up

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link add veth-h1 type veth peer name veth-r1
term1
lekhya@Alekhya:~$ sudo ip netns add host1
udo ip netns add router
udo ip netns add host2
sudo: authenticate] Password:
lekhya@Alekhya:~$ sudo ip netns add host1
udo ip netns add router
udo ip netns add host2
annot create namespace file "/run/netns/host1": File exists
annot create namespace file "/run/netns/router": File exists
annot create namespace file "/run/netns/host2": File exists
lekhya@Alekhya:~$ sudo ip link add veth-h1 type veth peer name veth-r1
udo ip link add veth-r2 type veth peer name veth-h2
lekhya@Alekhya:~$ sudo ip link set veth-h1 netns host1
udo ip link set veth-r1 netns router
udo ip link set veth-r2 netns router
udo ip link set veth-h2 netns host2
lekhya@Alekhya:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
udo ip netns exec host1 ip link set veth-h1 up
udo ip netns exec host1 ip link set lo up
lekhya@Alekhya:~$
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
```

router:

sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns del host1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link del veth-h2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns list
host2
router
host1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
Cannot find device "veth-h1"
Cannot find device "veth-h1"
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns list
host2
router
host1
```

host2:

sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT gro
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link add veth-h1 type veth peer name veth-r1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link add veth-r2 type veth peer name veth-h2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ip link show
```

STAGE 4 — Enable routing

sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ip addr
```

STAGE 5 — Test connectivity

sudo ip netns exec host1 ping -c 3 10.0.2.2

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.475 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.087 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.234 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2036ms
rtt min/avg/max/mdev = 0.087/0.265/0.475/0.159 ms
```

You must see:

3 packets transmitted, 3 received, 0% packet loss

STAGE 6 — Add delay and loss

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.475 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.087 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.234 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2036ms
rtt min/avg/max/mdev = 0.087/0.265/0.475/0.159 ms
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router tc qdisc replace dev veth-r2 root netem delay 20ms loss 1%
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 15074435788272599323
```

Verify:

sudo ip netns exec router tc qdisc show dev veth-r2

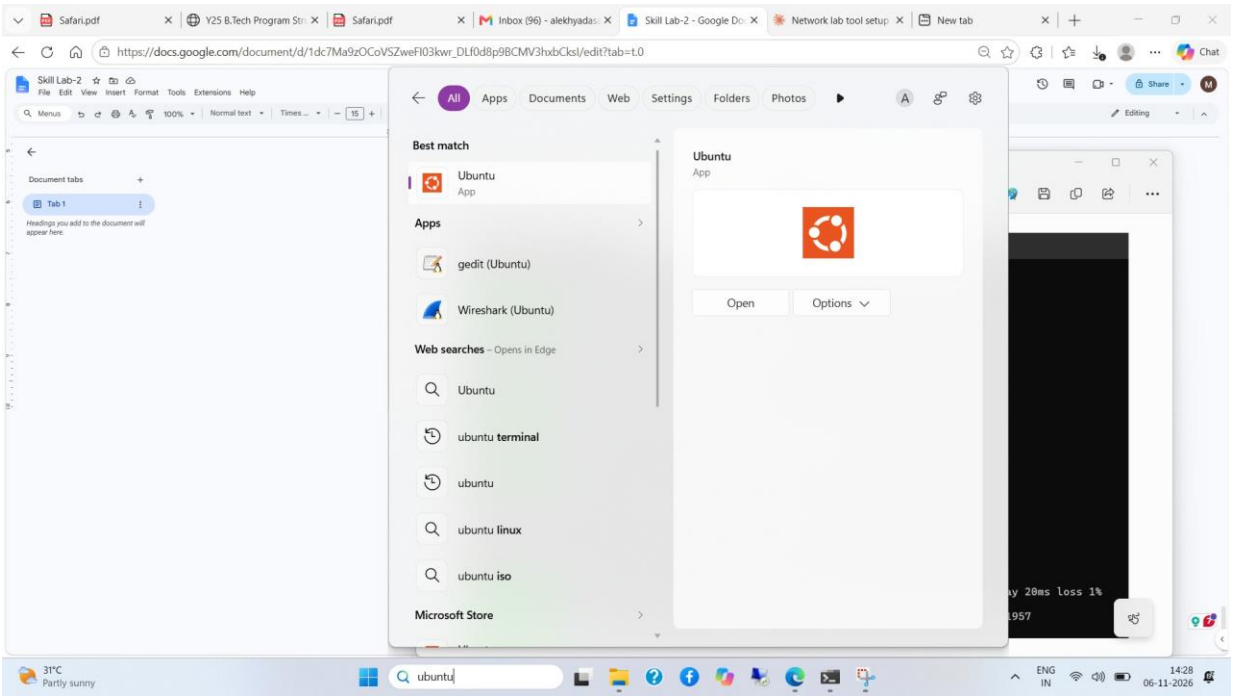
```
--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2036ms
rtt min/avg/max/mdev = 0.087/0.265/0.475/0.159 ms
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router tc qdisc replace dev veth-r2 root netem delay 20ms loss 1%
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 15074435788272599323
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ping -c 10 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
```

You must see netem delay 20ms loss 1%

STAGE 7 — Open 4 terminals

You need 4 terminals open now.

Terminal 1 — already open (used above)
Terminal 2 — open new (By going to Ubuntu option)



Terminal 3 — open new (By going to Ubuntu option)
Terminal 4 — open new (By going to Ubuntu option)

STAGE 8 — Start three captures simultaneously

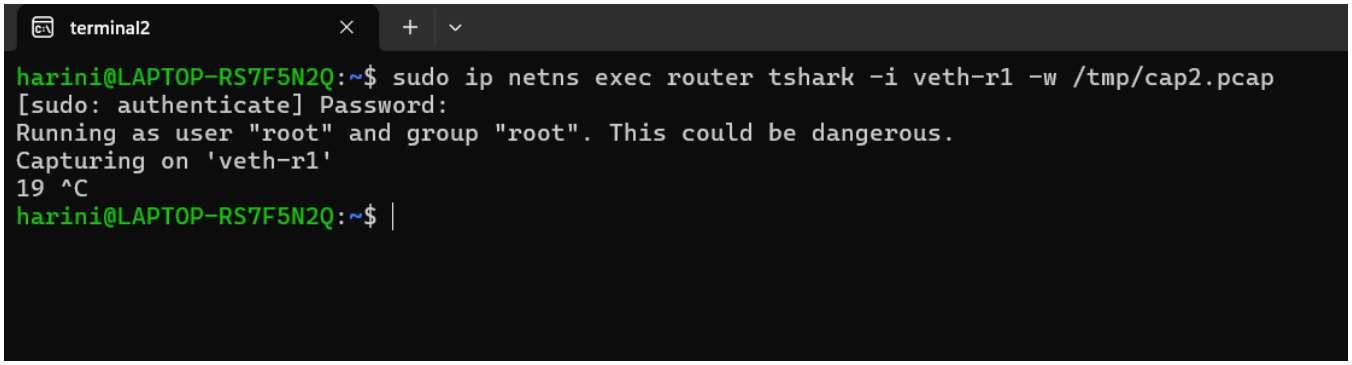
Terminal 1 — capture at host1

sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap

```
rtt min/avg/max/mdev = 20.248/20.604/21.943/0.491 ms
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
exec of "tshark" failed: No such file or directory
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
19 ^C
```

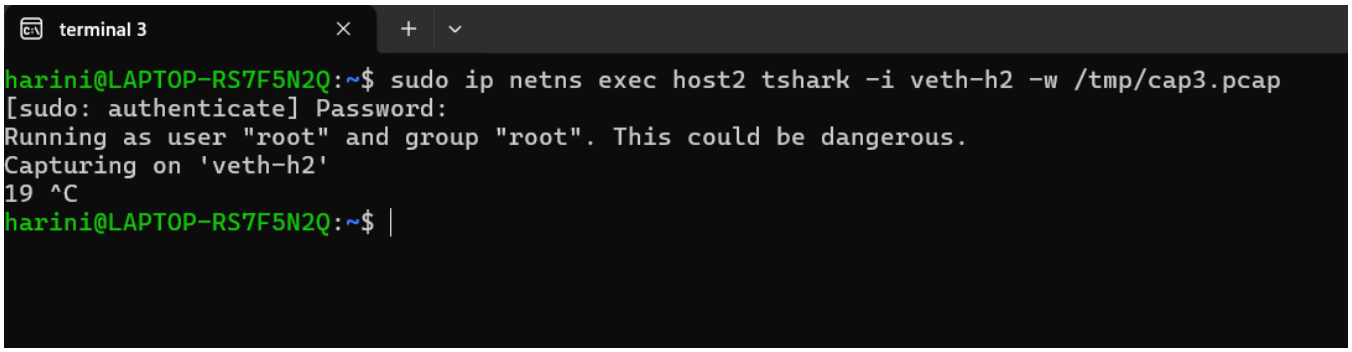
Terminal 2 — capture at router

sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap



Terminal 3 — capture at host2

sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap



All three show:

Capturing on 'veth-h1'

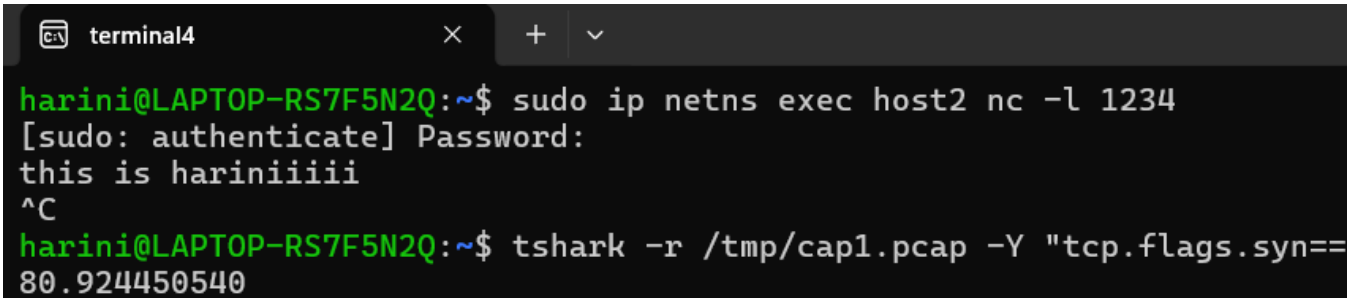
Leave all three running.

STAGE 9 — Send real message from host1 to host2

Go to Terminal 4.

First start the receiver on host2:

sudo ip netns exec host2 nc -l 1234



Open a **5th terminal.**

Connect from host1:

sudo ip netns exec host1 nc 10.0.2.2 1234

Now type this message and press Enter:

Hello from host1!

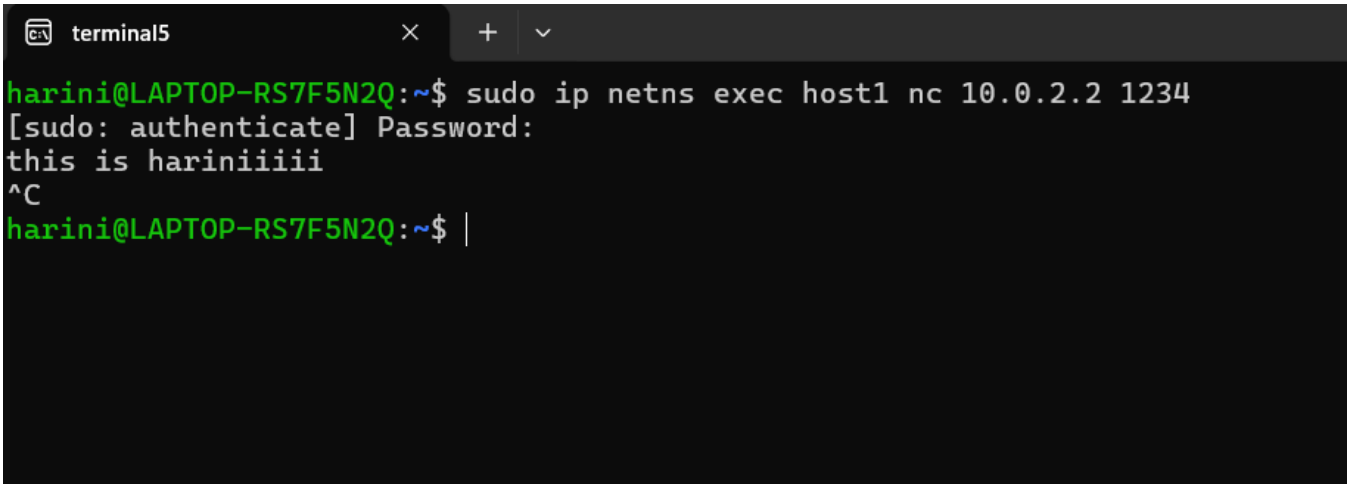
Type more messages:

This is Alekhya

Computer Networks Lab

Three Point Capture Experiment

Message traveling through router



STAGE 10 — What you see in real time

Terminal 4 (host2) shows your message arriving:

Hello from host1!

This is Alekhya

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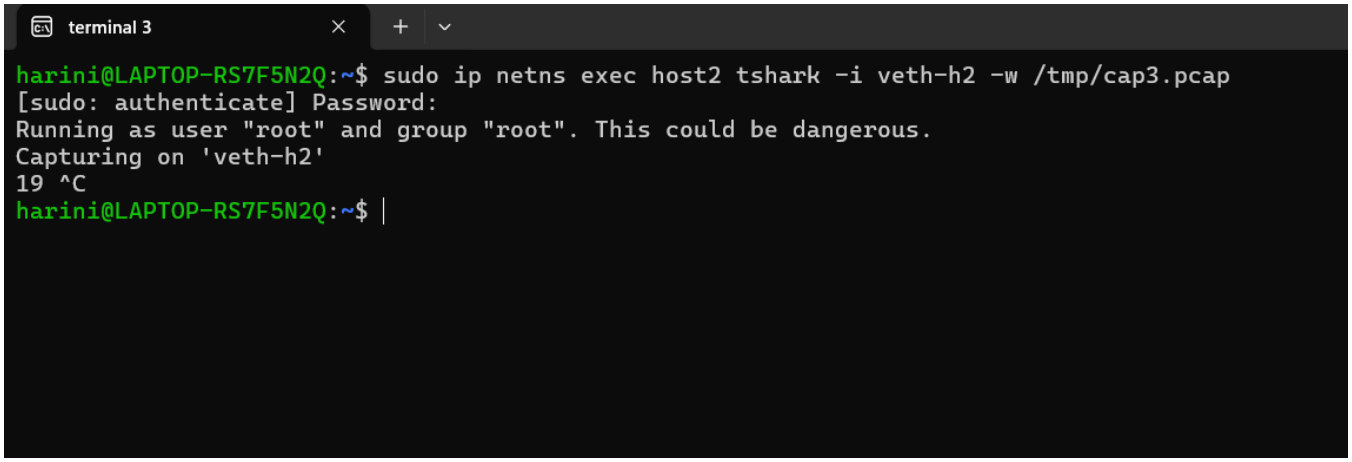
Three Point Capture Experiment

Message travelling through router

Terminal 1, 2, 3 (tshark) show packet counts increasing:

Capturing on 'veth-h1'

10 packets



STAGE 12 — Fix permissions

sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap

STAGE 13 — Check TTL

tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

Expected:

cap1 → 64

cap2 → 64

cap3 → 63

```
harini@LAPTOP-RS7F5N2Q:~$ ls -lh /tmp/cap*.pcap
-rw------- 1 root root 2.1K Nov 12 05:53 /tmp/cap1.pcap
-rw------- 1 root root 2.1K Nov 12 05:53 /tmp/cap2.pcap
-rw------- 1 root root 2.1K Nov 12 05:53 /tmp/cap3.pcap
harini@LAPTOP-RS7F5N2Q:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
harini@LAPTOP-RS7F5N2Q:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      4 63
      5 64
harini@LAPTOP-RS7F5N2Q:~$ tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      4 63
      5 64
harini@LAPTOP-RS7F5N2Q:~$ tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      5 63
      4 64
```

STAGE 14 — Read your actual message from the capture

tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e data.text 2>/dev/null

You will see your actual message text extracted from the packet:

Hello from host1!

This is Alekhya

Computer Networks Lab

Three Point Capture Experiment

Message travelling through router

```
      5 63
      4 64
harini@LAPTOP-RS7F5N2Q:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e data.text 2>/dev/null



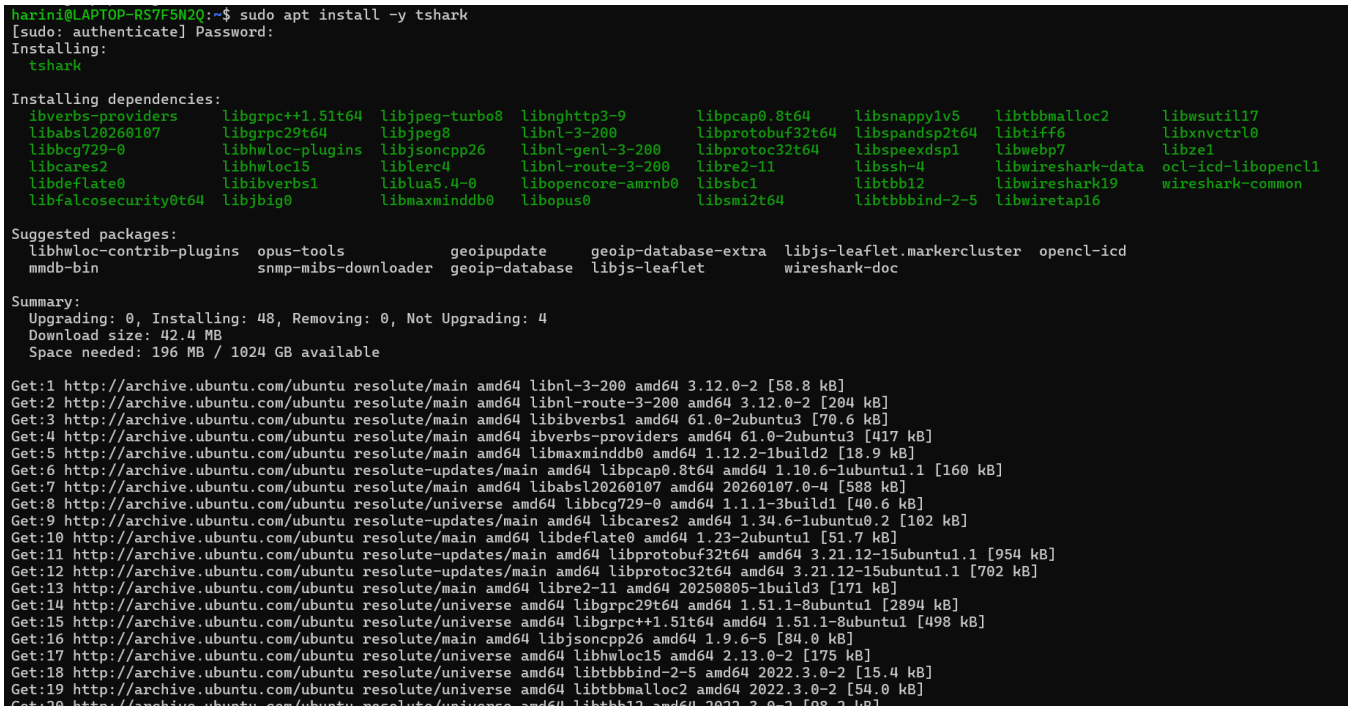

harini@LAPTOP-RS7F5N2Q:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234 && tcp.len > 0" -T fields -e tcp.payload
7468697320697320686172696e69696969690a
harini@LAPTOP-RS7F5N2Q:~$
```

Connect from host1:

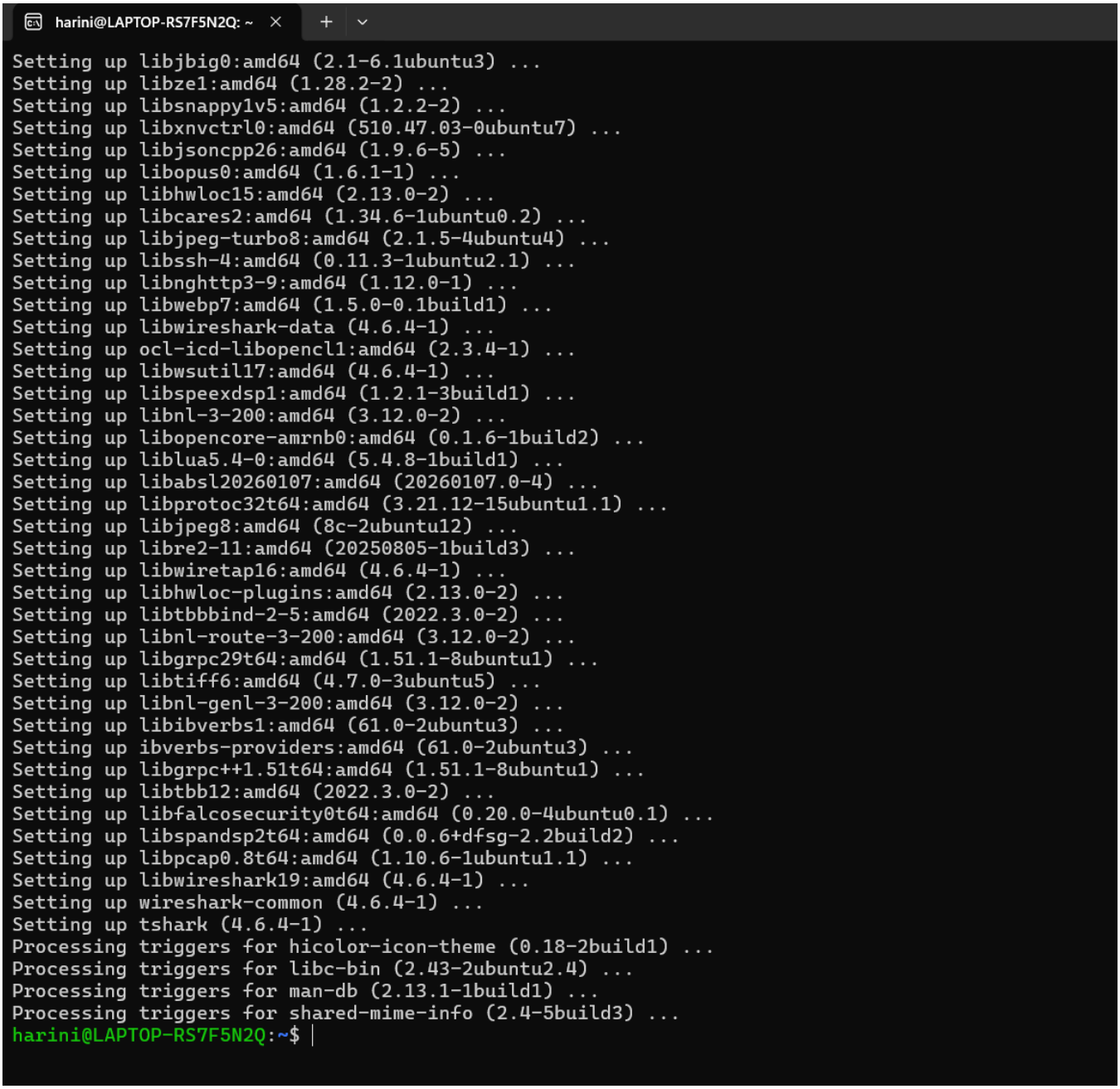
TERMINAL 1 — Setup (run all commands here)

Install tools:

sudo apt install -y tshark iproute2 netcat-openbsd wireshark



When asked → choose **Yes**



Create 3 nodes:

sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2

Verify:

sudo ip netns list

```
Processing triggers for shared-mime-info (2.4-5build3) ...
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns add host1 sudo ip netns add router sudo ip netns add host2
[sudo: authenticate] Password:
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns list
host1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns add router
harini@LAPTOP-RS7F5N2Q:~$  sudo ip netns list
router
host1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns add host2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns list
host2
router
host1
harini@LAPTOP-RS7F5N2Q:~$
```

You must see: host1 router host2

Create cables:

sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns add host1 sudo ip netns add router sudo ip netns add host2
[sudo: authenticate] Password:
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns list
host1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns add router
harini@LAPTOP-RS7F5N2Q:~$  sudo ip netns list
router
host1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns add host2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns list
host2
router
host1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link add veth-h1 type veth peer name veth-r1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link add veth-r2 type veth peer name veth-h2
harini@LAPTOP-RS7F5N2Q:~$
```

Plug cables into nodes:

sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link set veth-h1 netns host1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link set veth-r1 netns router
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link set veth-r2 netns router
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link set veth-h2 netns host2
harini@LAPTOP-RS7F5N2Q:~$
```

Configure host1:

sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link set veth-h1 netns host1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link set veth-r1 netns router
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link set veth-r2 netns router
harini@LAPTOP-RS7F5N2Q:~$ sudo ip link set veth-h2 netns host2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ip link set veth-h1 up
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 ip link set lo up
harini@LAPTOP-RS7F5N2Q:~$
```

Configure router:

sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up

```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router ip link set veth-r1 up
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router ip link set veth-r2 up
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec router ip link set lo up
harini@LAPTOP-RS7F5N2Q:~$
```

NOW USE ALL 4 TERMINALS

TERMINAL 1 — Capture at host1

sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap

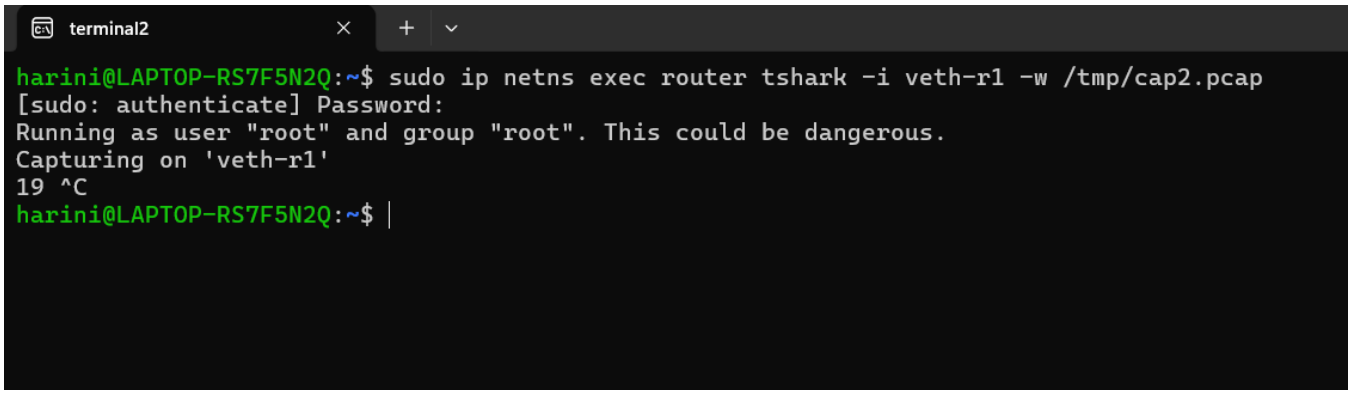
```
harini@LAPTOP-RS7F5N2Q:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap

[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
```

This terminal is left running.

TERMINAL 2 — Capture at router

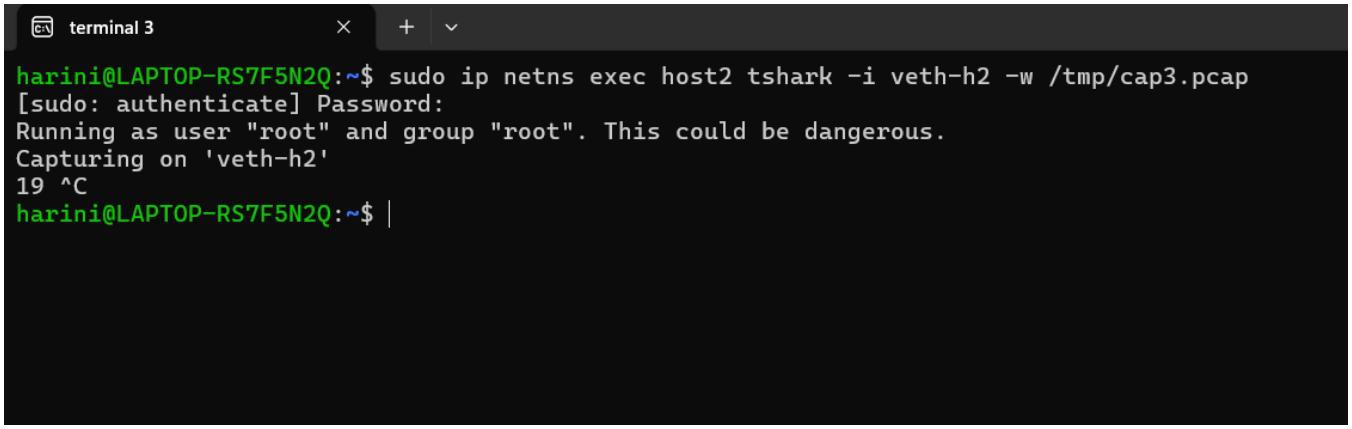
sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap



Leave running.

TERMINAL 3 — Capture at host2

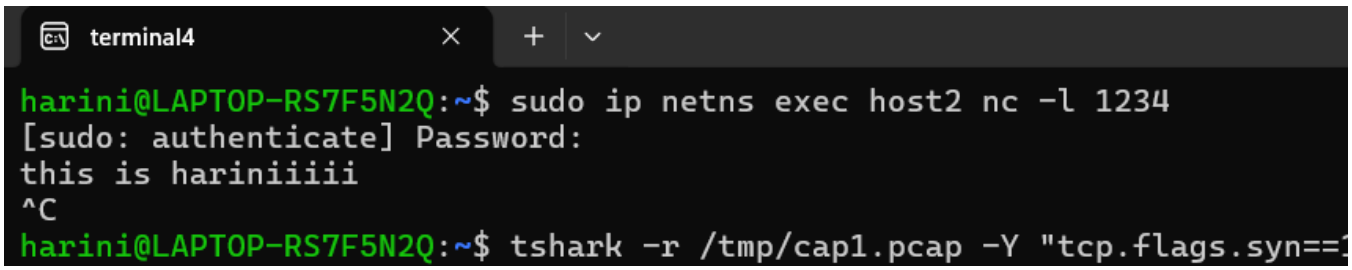
sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap



Leave running.

TERMINAL 4 — Send the message

Step 1 — Start receiver on host2:



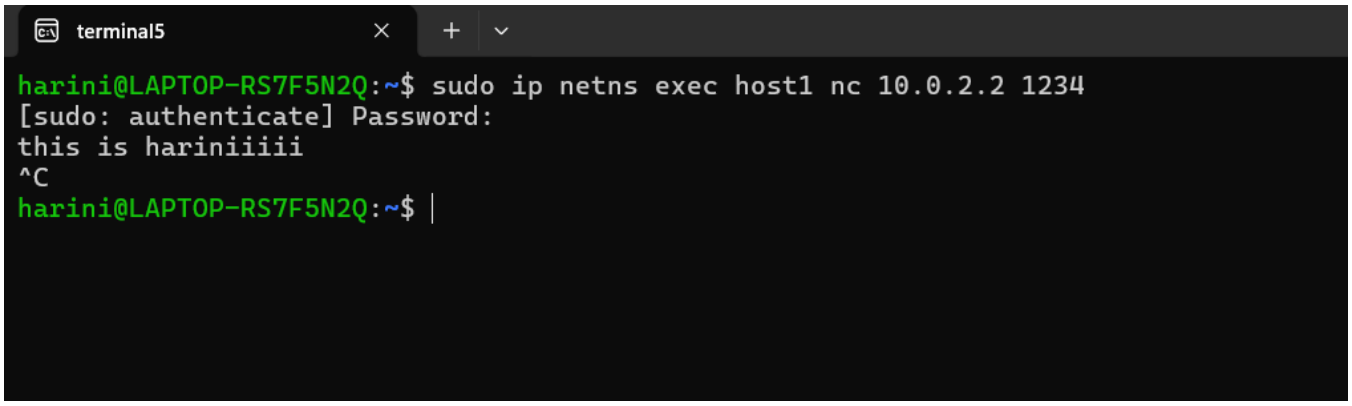
Cursor blinks. Leave it waiting.

Step 2 — Open a new Terminal 5

Step 3 — Connect from host1:

Paste the below command in terminal-5

sudo ip netns exec host1 nc 10.0.2.2 1234

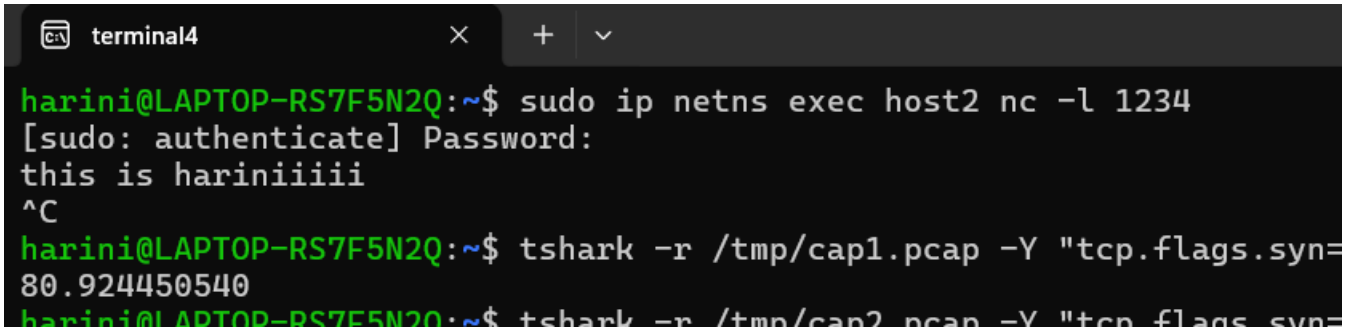


Type the below message in Terminal-5

This is Harinii

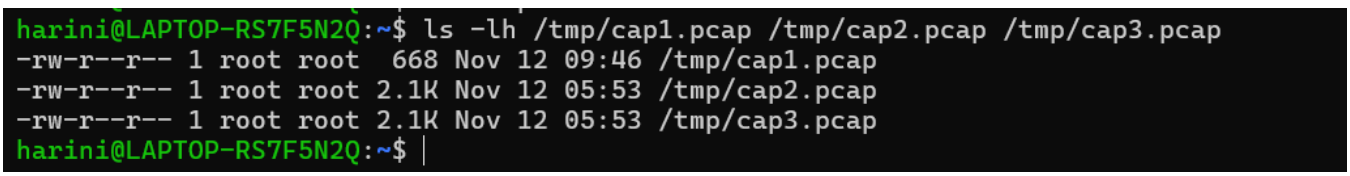
Step 5 — Check Terminal 4 first tab

You will see your message appearing there — that is host2 receiving it.



Check files are not empty:

ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap



Inside Wireshark:

Filter your message only:

tcp.port == 1234

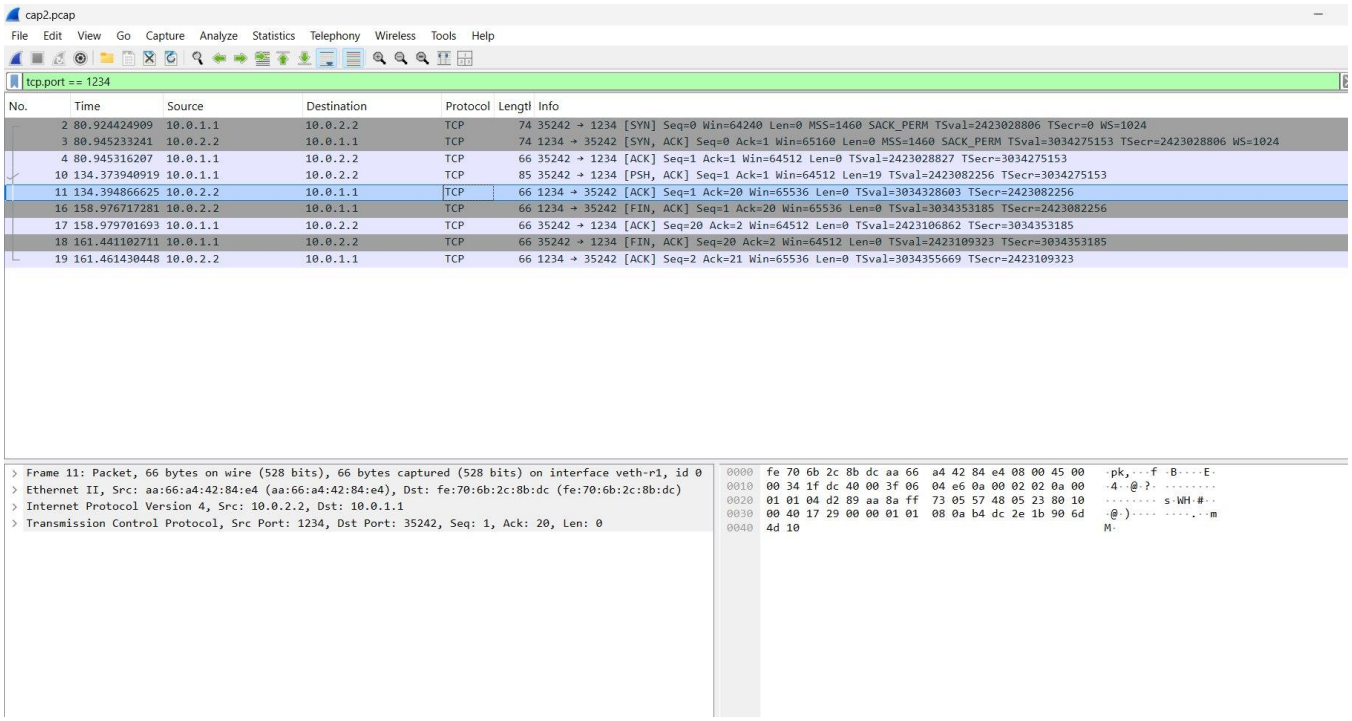
Click any packet → expand middle pane → you will see:

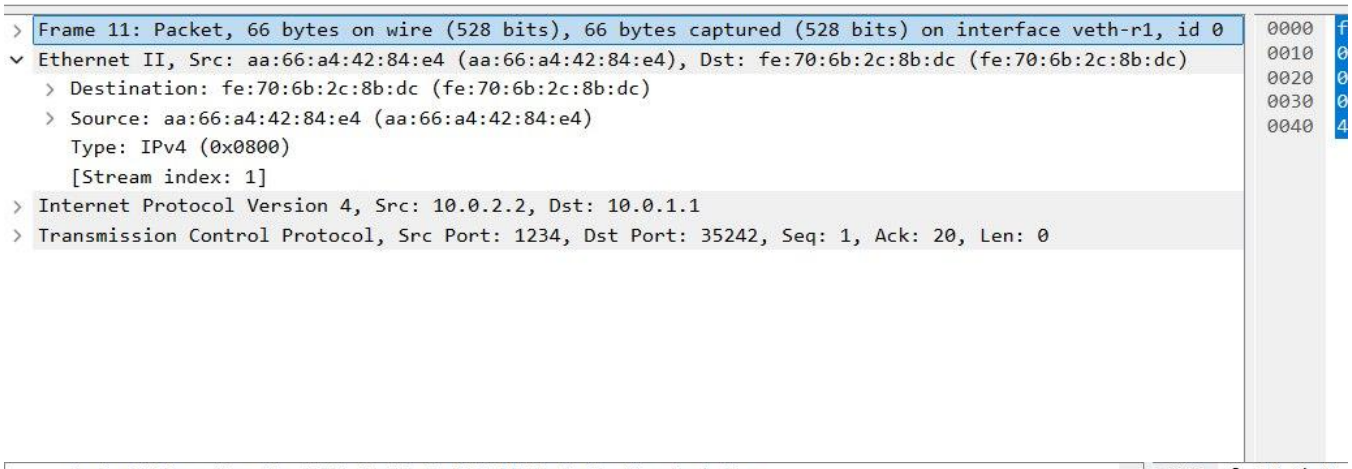
▼ Ethernet II

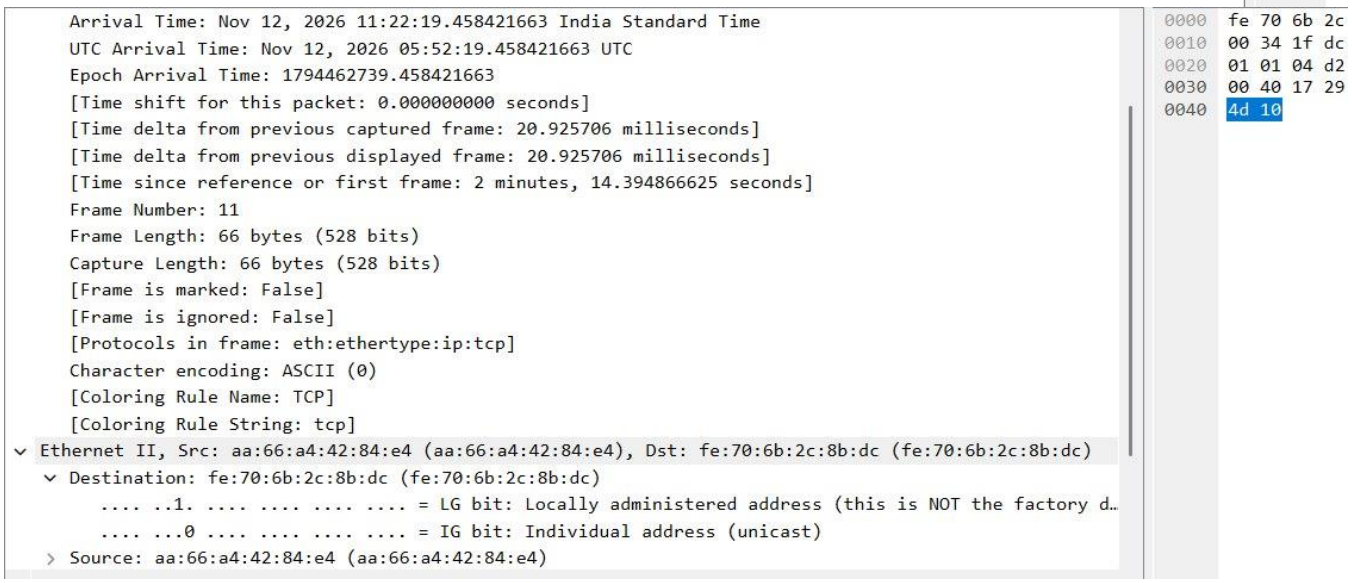
▼ Internet Protocol Version 4

Time to Live: 64 ← TTL visible here

▼ Transmission Control Protocol

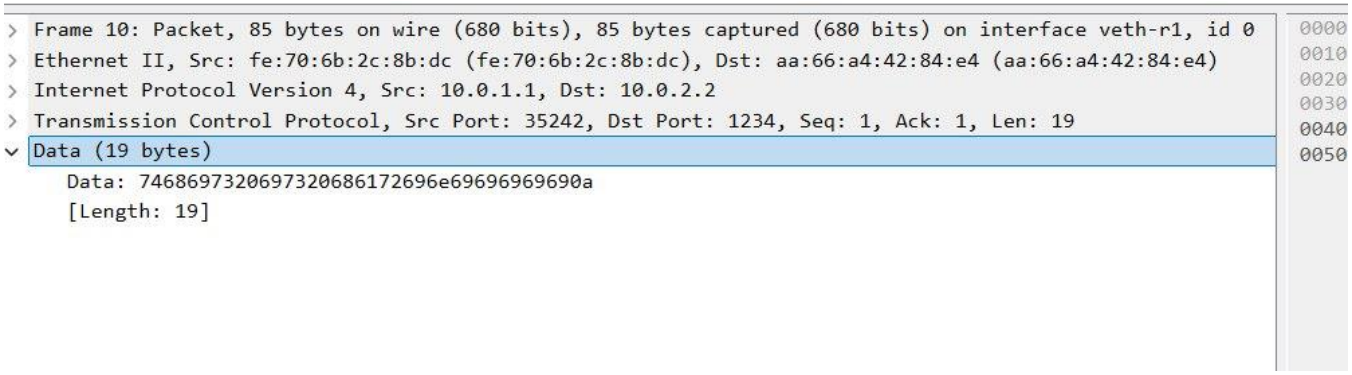






▼ Data

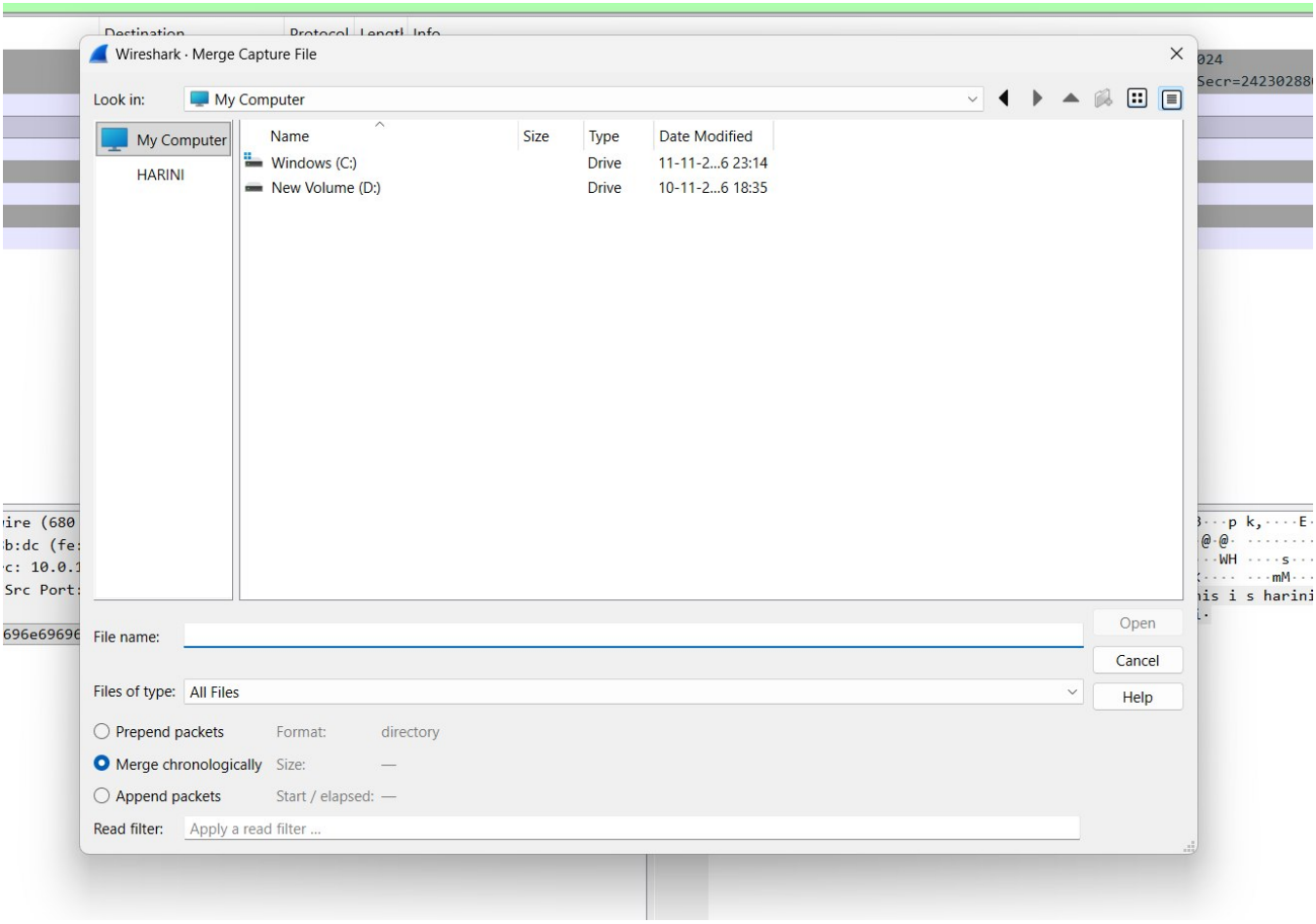
Hello from host1! ← your actual message visible here



Merge all 3 captures:

File → Merge → select cap1.pcap

File → Merge → select cap3.pcap



Add TTL column:

Right-click column header

→ Column Preferences

→ click +

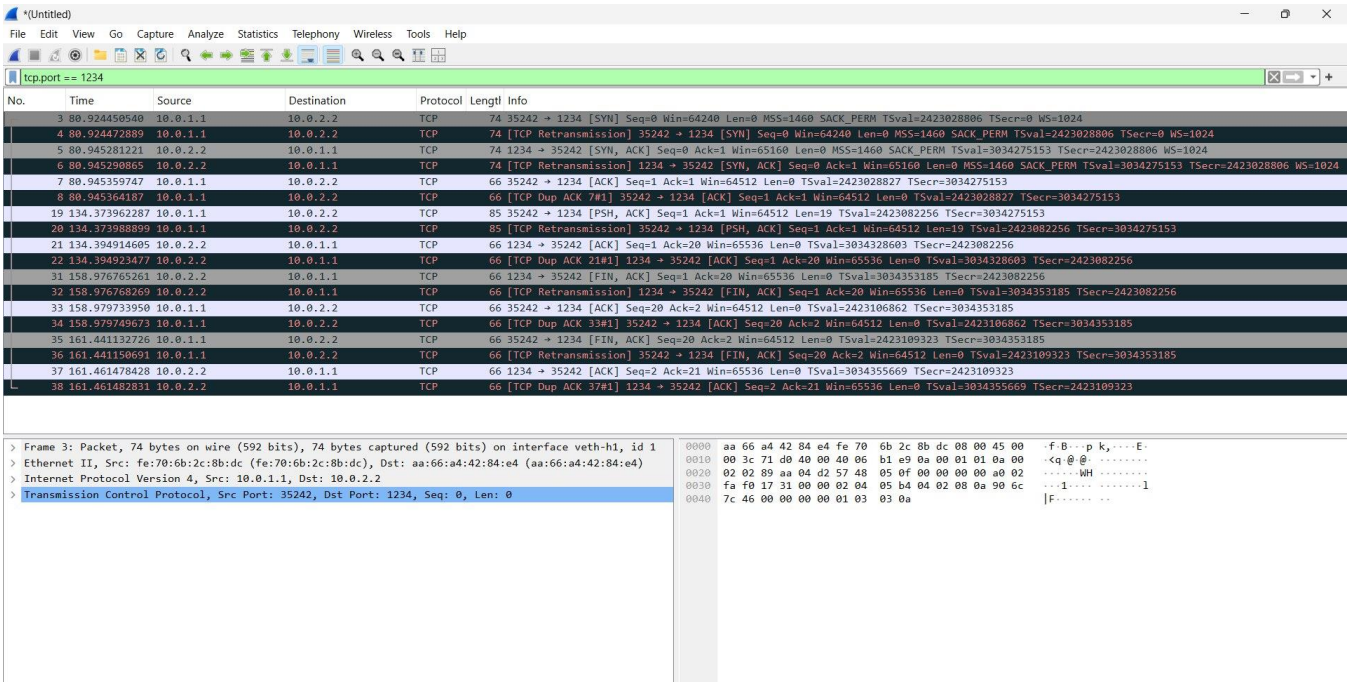
→ Title: TTL

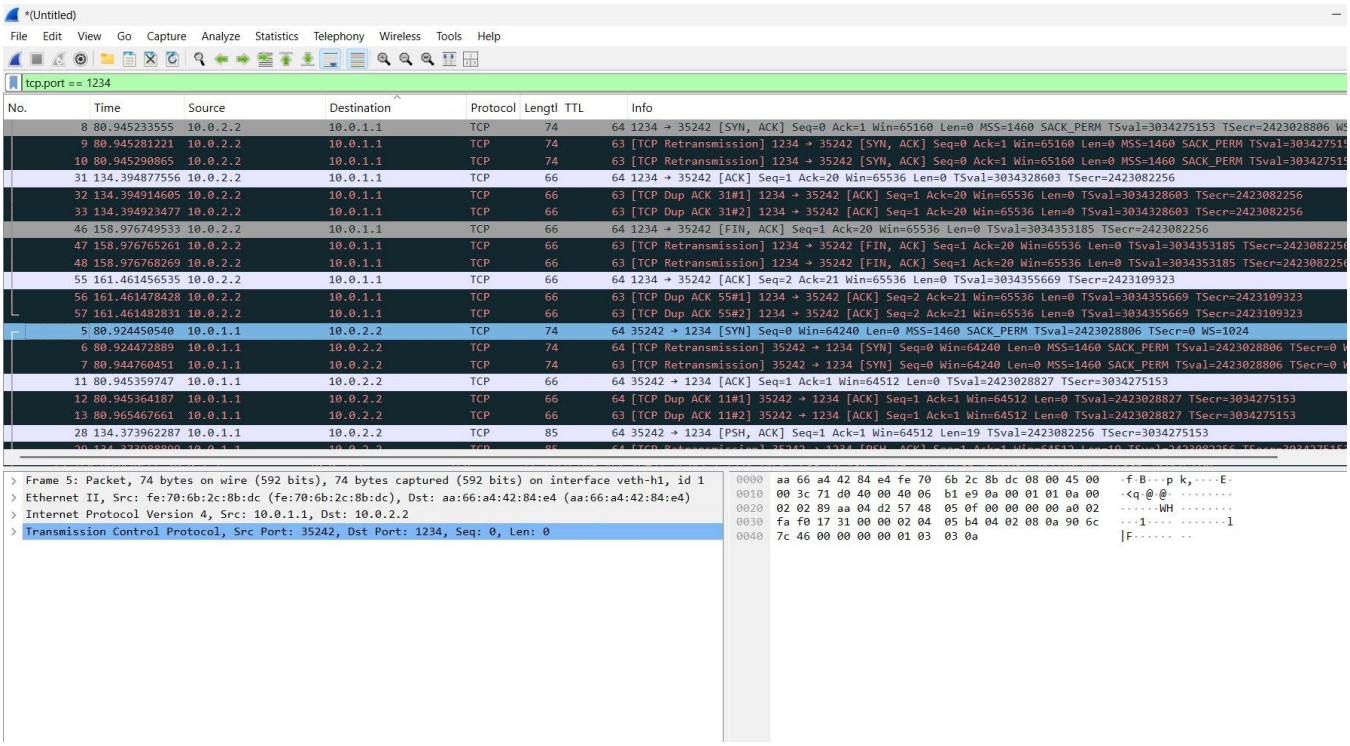
→ Type: Custom

→ Fields: ip.ttl

→ OK

Students can now see TTL = 64 at points (1) and (2) and TTL = 63 at point (3)





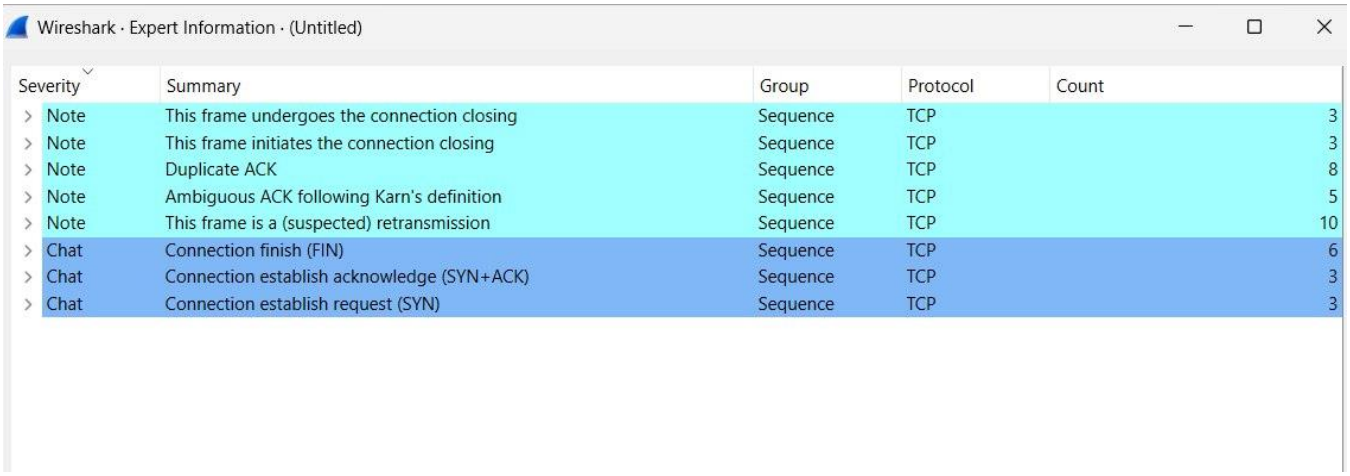
Show packet loss:

Analyze → Expert Information

CLEAN UP — Terminal 4

sudo ip netns del host1

sudo ip netns del router



Severity	Summary	Group	Protocol	Count
Note	This frame undergoes the connection closing	Sequence	TCP	3
Note	This frame initiates the connection closing	Sequence	TCP	3
Note	Duplicate ACK	Sequence	TCP	8
Note	Ambiguous ACK following Karn's definition	Sequence	TCP	5
Note	This frame is a (suspected) retransmission	Sequence	TCP	10
Chat	Connection finish (FIN)	Sequence	TCP	6
Chat	Connection establish acknowledge (SYN+ACK)	Sequence	TCP	3
Chat	Connection establish request (SYN)	Sequence	TCP	3

sudo ip netns del host2

Verify:

sudo ip netns list

Must return blank.